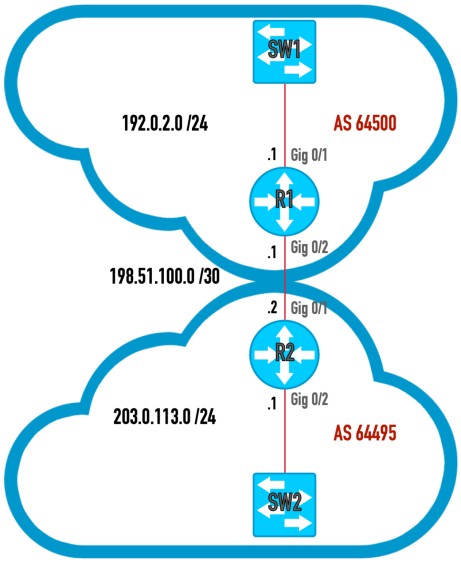
BGP

Topology



Initial Configuration Commands

R1:

```
enable
conf t
host R1
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
```

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```
no ip domain-lookup
int gig 0/1
ip address 192.0.2.1 255.255.255.0
no shut
int gig 0/2
ip address 198.51.100.1 255.255.255.252
no shut
end
copy run star
```

R2:

```
enable
conf t
host R2
no banner motd
no banner login
no banner exec
no banner incoming
line vty 0 15
password cisco
login
exec-timeout 0 0
line con 0
logging synchronous
exit
no ip domain-lookup
int gig 0/1
ip address 198.51.100.2 255.255.255.252
no shut
int gig 0/2
ip address 203.0.113.1 255.255.255.0
no shut
router ospf 1
network 0.0.0.0 0.0.0.0 area 0
end
copy run star
```

Lab Tasks

1. On R1, let's start up the BGP routing process globally as well as specify the autonomous system number to which this router belongs.
2. Next, let's specify our BGP neighbor, as well what autonomous system our neighbor belongs to.
3. Lastly, let's advertise our internal network over to our neighbor (R2).

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4. On R2, let's set up the BGP routing process, designate R1 as our neighbor, and advertise our internal network to R1.
5. On both R1 and R2, let's see what networks each router has knowledge of in their BGP tables.

Solution

Step 1: On R1, let's start up the BGP routing process globally as well as specify the autonomous system number to which this router belongs.

```
R1>en
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#router bgp 64500

(##As you can see here, R1 belongs to the autonomous system of
64500.)
```

Step 2: Next, let's specify our BGP neighbor, as well what autonomous system our neighbor belongs to.

```
R1(config-router)#neighbor 198.51.100.2 remote-as 64495

(##BGP does not dynamically learn its neighbors, so we have to
specifically identify those neighbors.  As you can see here, we
are identifying Gig 0/1 on R2 as our neighbor.)
```

Step 3: Next, let's advertise our internal network over to our neighbor (R2).

```
R1(config-router)#network 192.0.2.0 mask 255.255.255.0
R1(config-router)#end
```

Step 4: On R2, let's set up the BGP routing process, designate R1 as our neighbor, and advertise our internal network to R1.

```
R2>en
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#router bgp 64495
R2(config-router)#neighbor 198.51.100.1 remote-as 64500
R2(config-router)#network 203.0.113.0 mask 255.255.255.0
R2(config-router)#end
```

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Step 5: On both R1 and R2, let's see what networks each router has knowledge of in their BGP tables.

```
R1#show ip bgp
BGP table version is 3, local router ID is 198.51.100.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   192.0.2.0        0.0.0.0                  0         32768 i
 *>   203.0.113.0      198.51.100.2             0             0 64495 i
```

```
R2#show ip bgp
BGP table version is 3, local router ID is 203.0.113.1
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,
              x best-external, a additional-path, c RIB-compressed,
              t secondary path,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

     Network          Next Hop            Metric LocPrf Weight Path
 *>   192.0.2.0        198.51.100.1             0             0 64500 i
 *>   203.0.113.0      0.0.0.0                  0         32768 i
```

(##Notice how R1 now has knowledge of the 203.0.113.0 /30 network because of the BGP routing process we created, and that network can be reached via 198.51.100.2.)

(##Notice how R2 now has knowledge of the 192.0.2.0 /24 network because of the BGP routing process we created, and that network can be reached via 198.51.100.1.)

(##Based on the BGP tables, we can see that the routes on each of the routers are valid, and they are the best routes to reach the network in the neighboring autonomous system.)

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